

Erdős Problem #665

EP-665

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

A pairwise balanced design for $\{1, \dots, n\}$ is a collection of sets $A_1, \dots, A_m \subseteq \{1, \dots, n\}$ such that $2 \leq |A_i| < n$ and every pair of distinct elements $x, y \in \{1, \dots, n\}$ is contained in exactly one A_i .

Is there a constant $C > 0$ and, for all large n , a pairwise balanced design such that $|A_i| > n^{1/2} - C$ for all $1 \leq i \leq m$?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>